

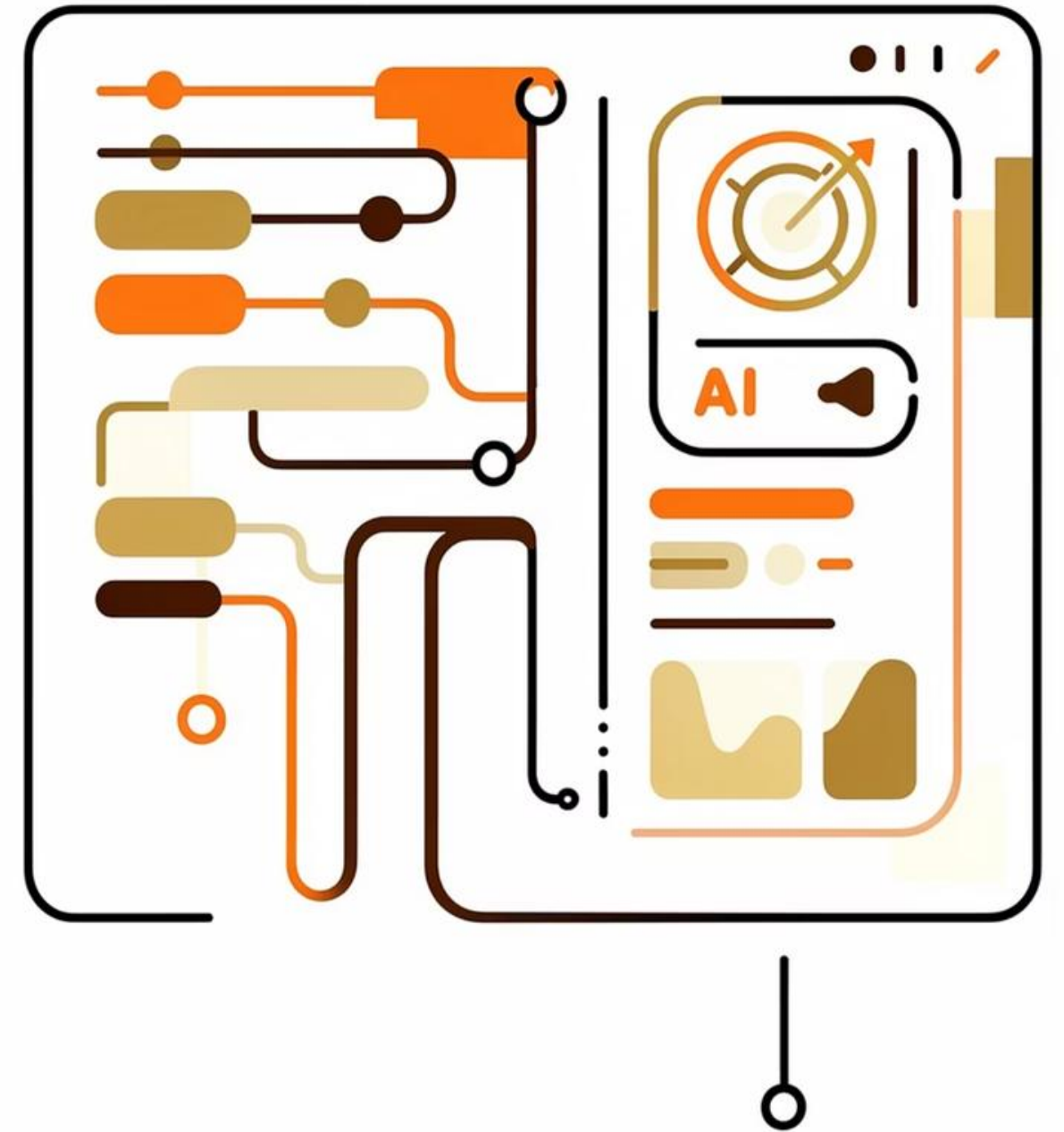
WATCHDOG

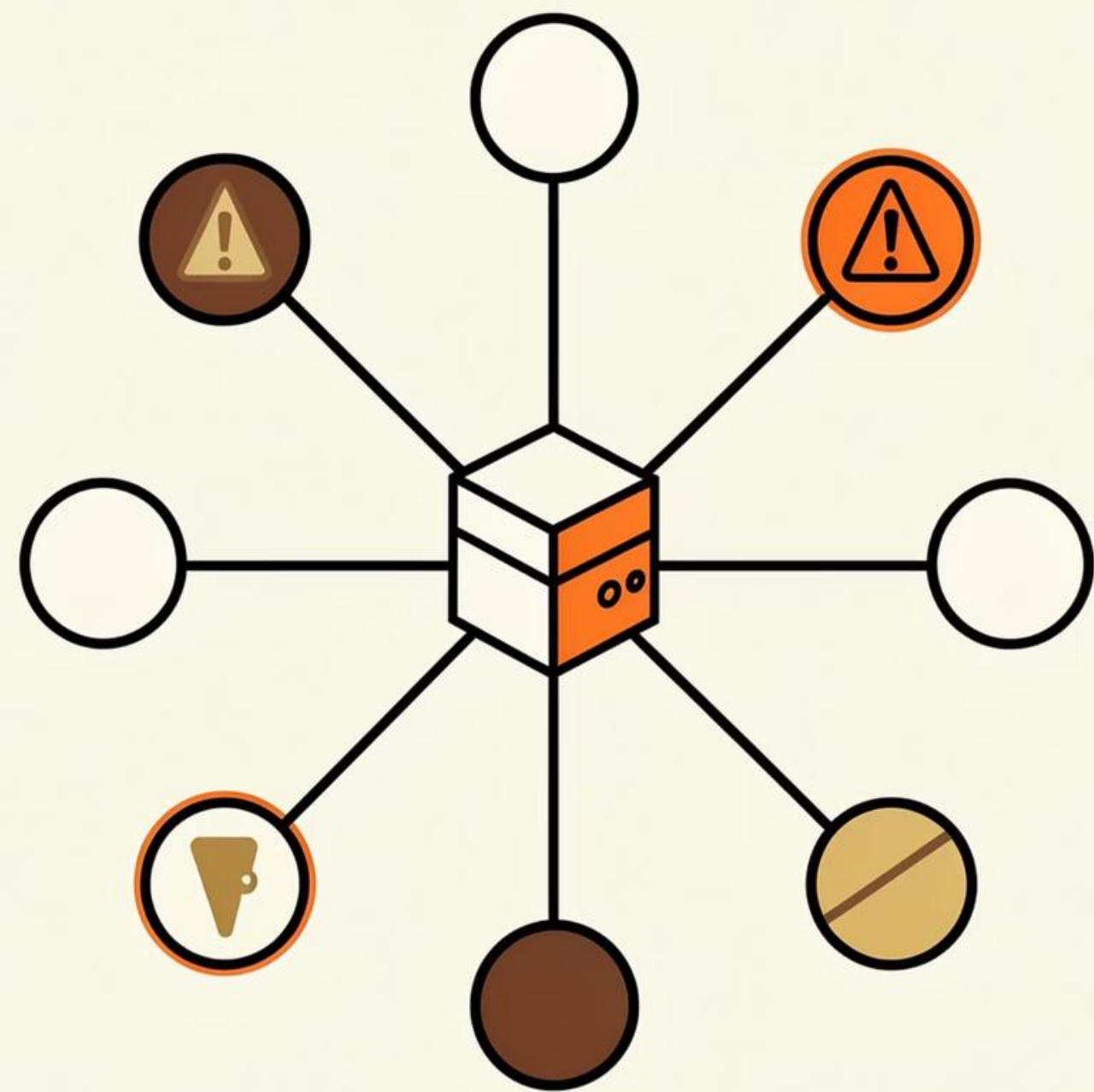
Intelligent Observability & Event Watchdog

AI-Powered SRE Monitoring & Incident Intelligence Platform

ENGINEERING LEADERSHIP REVIEW

ENTERPRISE INFRASTRUCTURE





THE OBSERVABILITY GAP IS WIDENING

Modern distributed systems generate billions of events daily — yet critical failures remain invisible until they cascade into outages.

Distributed Complexity

Microservices, containers, and cloud-native architectures obscure failure boundaries across dozens of components.

Hidden Failures

Silent degradation in latency, error rates, and resource exhaustion goes undetected without proactive instrumentation.

Delayed Detection

Mean time to detect (MTTD) remains unacceptably high — teams react to incidents rather than prevent them.

ONE PLATFORM. COMPLETE VISIBILITY.

WatchDog unifies log ingestion, anomaly detection, health scoring, and AI-driven insights into a single intelligent observability layer — purpose-built for SRE teams at scale.



Log Ingestion

High-throughput ingestion pipeline parsing structured and unstructured telemetry in real time.



Health Scoring

Composite service health scores aggregate latency, error rate, and availability signals.



AI Insights

LLM-powered root-cause analysis and incident summaries accelerate MTTR.



Anomaly Detection

Statistical and ML-based detection identifies deviations across metrics, logs, and traces.



Smart Alerts

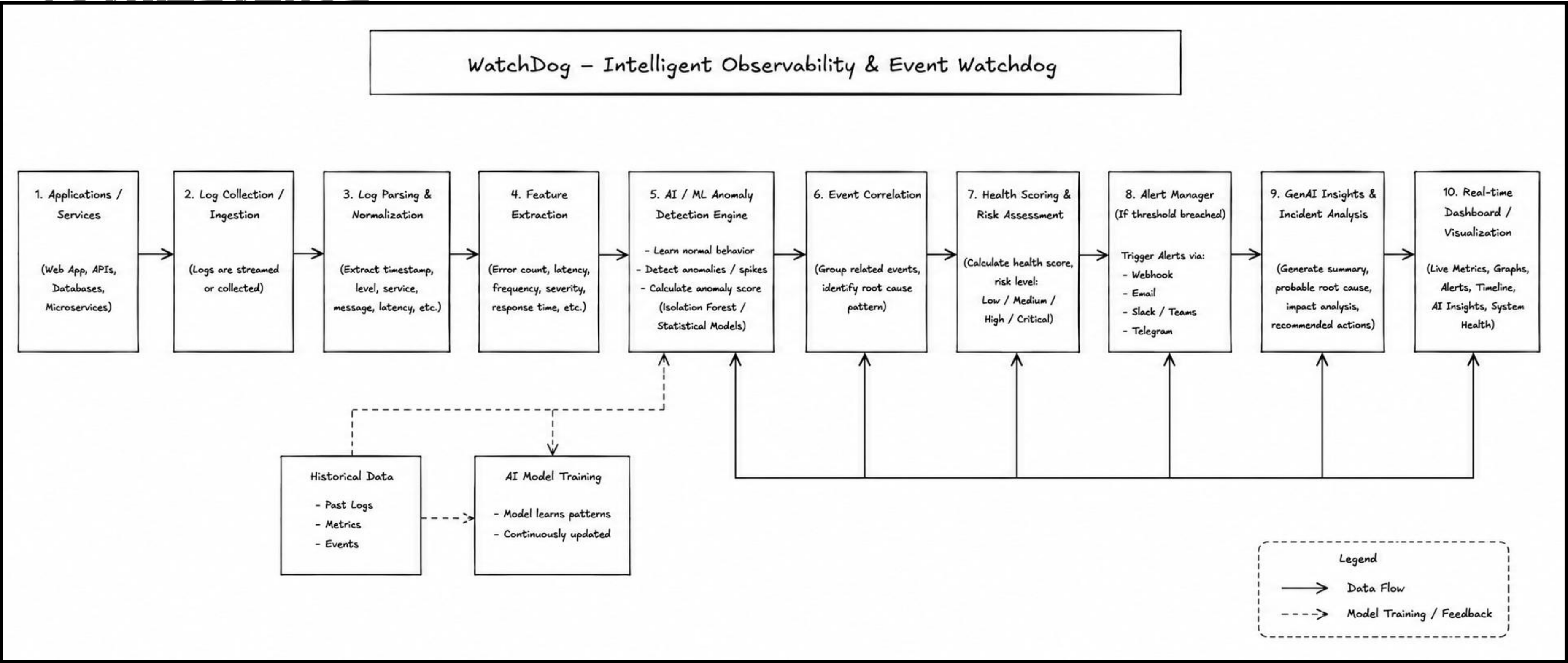
Contextual, deduplicated alerts reduce noise and surface only actionable incidents.



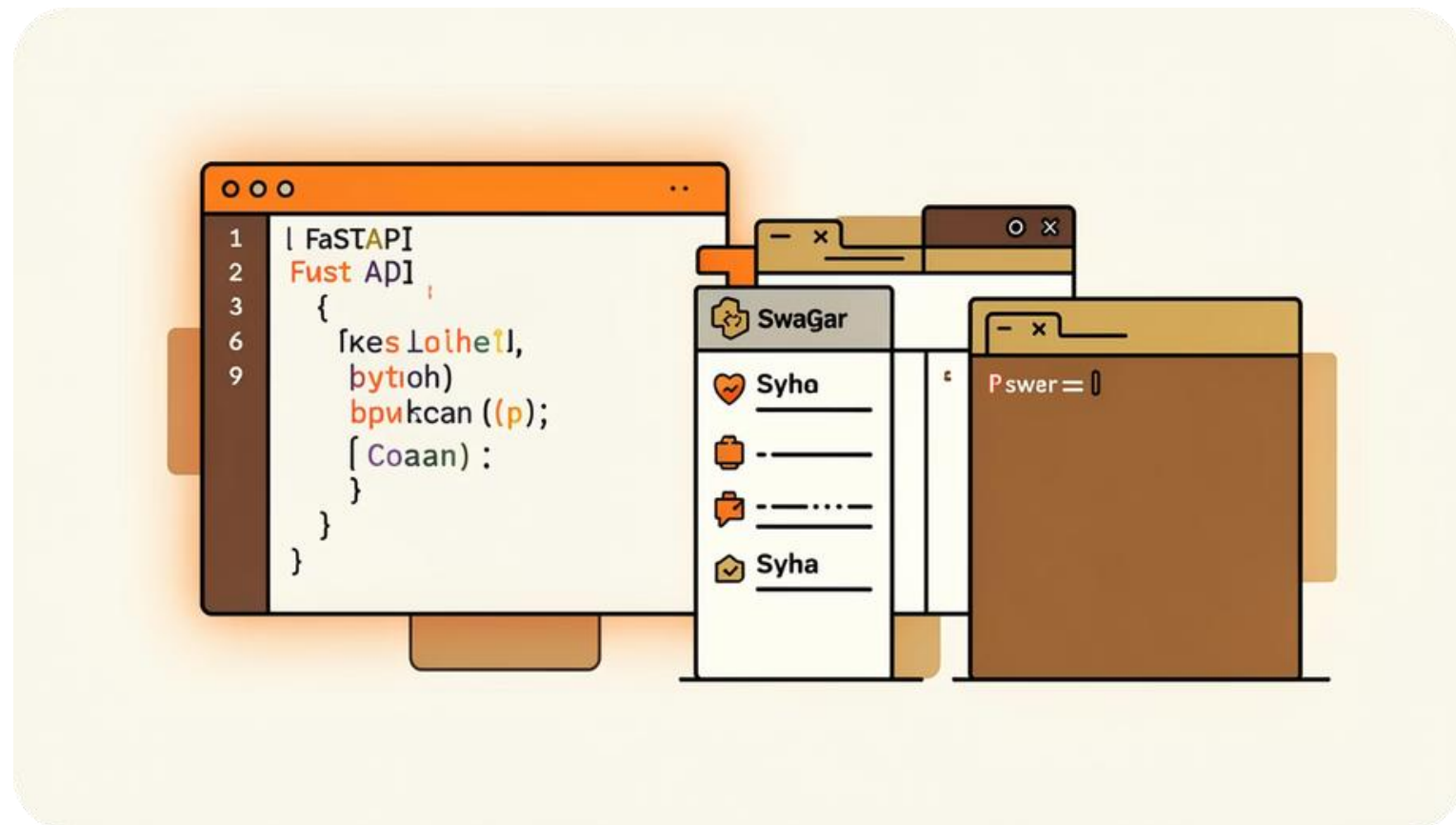
Unified Dashboard

Real-time visualization of system health, active alerts, and AI-generated recommendations.

SYSTEM



BACKEND ENGINEERING



WatchDog's backend is built on a modern, production-grade Python stack — engineered for reliability, extensibility, and developer experience.

FastAPI

Async Python framework powering high-performance REST APIs with auto-generated OpenAPI specs.

SQLite + SQLAlchemy

Lightweight persistence layer with ORM abstraction for rapid iteration and portability.

Observability Engine

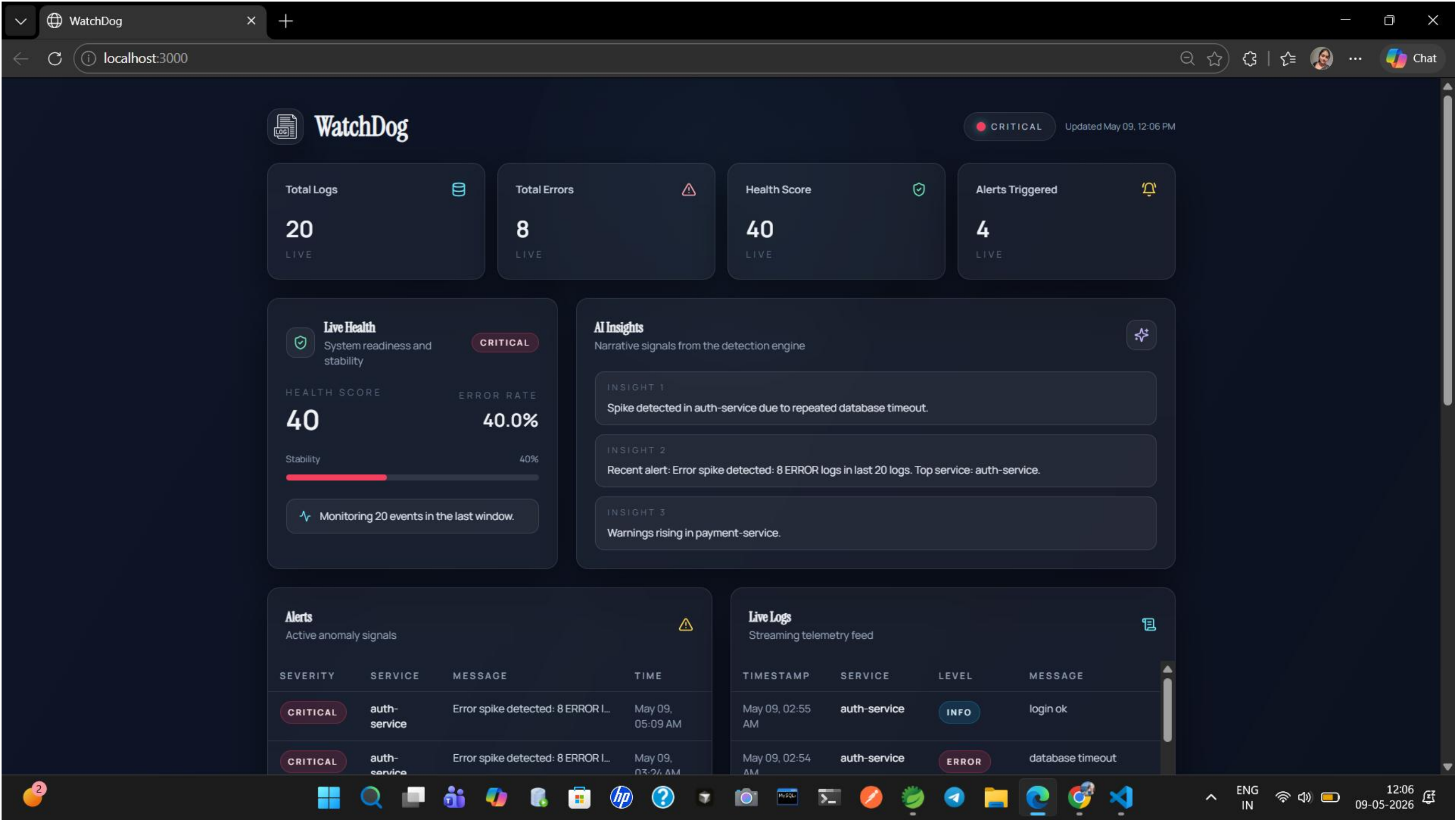
Core engine processes telemetry streams, computes health scores, and triggers alert rules.

Metrics & Alert APIs

Typed endpoints expose aggregated metrics, alert state, and AI insight payloads.

REAL-TIME DASHBOARD

A single pane of glass for SRE teams — live metrics, active alerts, AI insights, and log streams unified in a premium, dark-themed interface.



AI OBSERVABILITY FEATURES

WatchDog goes beyond traditional monitoring by embedding AI intelligence directly into the observability pipeline — transforming raw telemetry into actionable SRE insight.

Anomaly Detection

Detects statistical outliers in latency, throughput, and error rates across all services.

Health Scoring

Composite scores aggregate multiple signals into a single, actionable service health indicator.

AI-Generated Insights

LLM-powered summaries explain why an anomaly occurred and suggest remediation steps.

Incident Analysis

Correlates log patterns, metric spikes, and trace data to surface root causes automatically.

📄 **Example Insight:** "Spike detected in auth-service due to repeated database timeout. Connection pool exhausted at 14:32 UTC. Recommend increasing pool size and reviewing query latency on users table."

WHAT WE BUILT — AND WHAT'S

Delivered

- Full-stack observability platform from ingestion to insight
- Intelligent incident detection and correlation
- Real-time monitoring with sub-second alert latency
- AI-powered root-cause analysis and summarization

Roadmap

- WebSocket streaming for live log tail and metric push
- Kubernetes-native monitoring and auto-discovery
- ML-based anomaly detection with adaptive thresholds
- Slack integration and distributed tracing support

THANK YOU

WatchDog — Built for the complexity of modern infrastructure. Designed for the teams that keep it running.

WATCHDOG

INTELLIGENT OBSERVABILITY

AI-POWERED SRE